

Exercise Sheet 0

Question 1. Registration

Done.

Question 2. Comparing the efficiency of an algorithm

1. For A_1 :

$$t_1(n) = 10000 + 20 \cdot (\log_2 n)^2 = 10000 + 20 \cdot (\log_2 1024)^2 = 10000 + 20 \cdot 10^2 = 12000$$

For A_2 :

$$t_2(n) = 2 \cdot \sqrt{n} = 2 \cdot \sqrt{1024} = 2 \cdot 32 = 64$$

Since $64 < 12000$, A_2 is faster for $n = 1024$.

2. To find n where the two algorithms run at the same speed, solve $t_1(n) = t_2(n)$:

$$10000 + 20 \cdot (\log_2 n)^2 = 2 \cdot \sqrt{n}$$

Using Wolfram Alpha, the approximate solution is $n \approx 1.5e + 08$.

3. For large n , the growth rates dominate:

- $t_1(n)$ is dominated by $20 \cdot (\log_2 n)^2$, a **logarithmic-squared growth**.
- $t_2(n)$ is dominated by $2 \cdot \sqrt{n}$, a **square-root growth**.

Thus, as $n \rightarrow \infty$, $t_1(n)$ becomes much smaller than $t_2(n)$.

Conclusion: We should implement A_1 , as it is faster for large inputs.

Question 3. Small Riddles

1. We can find the heavier ball in 3 weighings by dividing and conquering:

- First Weighing: Divide 27 balls into three groups of 9. Weigh two groups against each other. If the scale tips, the heavier ball is in the tipped group. If the scale balances, the heavier ball is in the unweighed group.
- Second Weighing: Take the group of 9 containing the heavier ball and divide it into three groups of 3. Weigh two groups against each other. Again, the heavier ball is in the tipped group (or unweighed group if balanced).
- Third Weighing: Take the group of 3 containing the heavier ball and weigh two of them against each other. If the scale tips, the heavier one is the answer. If the scale balances, the unweighed ball is the heavier one.

2. It is not possible to cover the modified chessboard with 31 dominoes.

- A standard 8×8 chessboard has 32 black and 32 white squares. Removing two diagonally opposite corners (which are the same color, say black) leaves $32 - 2 = 30$ black squares and 32 white squares.
- A 2×1 domino always covers one black and one white square. Thus, 31 dominoes would cover 31 black and 31 white squares. But our modified board has 30 black and 32 white squares—an imbalance that makes full coverage impossible.